

Manufacturing Quality Control & Predictive Maintenance

Project Report

An End-to-End Data Analytics, Exploratory Data Analysis (EDA),
and Machine Learning Solution for Manufacturing Quality Control
and Predictive Maintenance

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Tools Used

Python | NumPy | SQL | Pandas | Matplotlib
Plotly | Scikit-learn | Streamlit | Joblib

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Executive Summary

The manufacturing industry relies heavily on the uninterrupted performance of machines and consistent product quality to achieve operational excellence. Unexpected machine failures, unplanned production downtime, and quality defects can significantly increase maintenance costs, reduce productivity, and impact customer satisfaction. To address these challenges, this project presents an end-to-end **Manufacturing Quality Control & Predictive Maintenance** solution that combines data analytics, exploratory data analysis (EDA), machine learning, and an interactive web application.

The project began with the collection and preparation of manufacturing data from multiple business entities, including machine information, production batches, sensor readings, maintenance logs, quality inspections, failure events, supplier information, operator information, and downtime records. The data was cleaned by handling missing values, correcting inconsistent records, validating data types, and identifying outliers to ensure reliable analysis.

Exploratory Data Analysis (EDA) was performed to uncover operational patterns and business insights related to machine performance, production quality, maintenance activities, downtime events, and sensor behavior. These insights helped identify the factors that contribute to machine failures and manufacturing inefficiencies, enabling a better understanding of operational risks.

To support predictive maintenance, a **Random Forest Classifier** was developed to predict whether a machine is likely to fail within the next **seven days** based on real-time sensor measurements, including temperature, vibration, pressure, oil level, voltage, current, RPM, sound level, and machine health score. The model was evaluated using standard classification metrics, including **Accuracy, Precision, Recall, and F1-Score**, to assess its predictive performance.

Finally, the trained model was deployed as an interactive **Streamlit** web application, allowing users to input machine sensor readings and receive instant fai

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Business Problem

Introduction

The manufacturing industry is one of the largest contributors to global economic growth, where maintaining high product quality and uninterrupted production is essential for achieving operational excellence. Modern manufacturing facilities depend on advanced machinery, automated production lines, and continuous monitoring systems to meet production targets while maintaining strict quality standards.

However, unexpected machine failures, equipment degradation, and inconsistent product quality remain significant challenges. These issues often lead to production delays, increased maintenance costs, higher defect rates, and reduced customer satisfaction. As manufacturing operations become increasingly data-driven, organizations generate vast amounts of operational data from machines, sensors, quality inspections, production lines, maintenance activities, suppliers, and inventory systems. Converting this raw data into meaningful insights has become a key business requirement.

Business Challenges

Manufacturing organizations commonly face several operational and strategic challenges that directly impact productivity and profitability.

Unplanned Machine Downtime

Unexpected equipment failures interrupt production schedules, resulting in lost productivity, delayed deliveries, and increased operational expenses.

Quality Control Issues

Variations in production processes may lead to defective products, increased rejection rates, customer complaints, and higher quality assurance costs.

Reactive Maintenance

Many organizations perform maintenance only after equipment fails, which increases repair costs and extends machine downtime.

Lack of Real-Time Insights

Although manufacturing systems collect large volumes of sensor and operational data, organizations often lack analytical tools to transform this information into actionable business insights.

Increasing Operational Costs

Machine failures, poor production efficiency, and maintenance delays contribute to higher manufacturing and operational expenses.

Project Goal

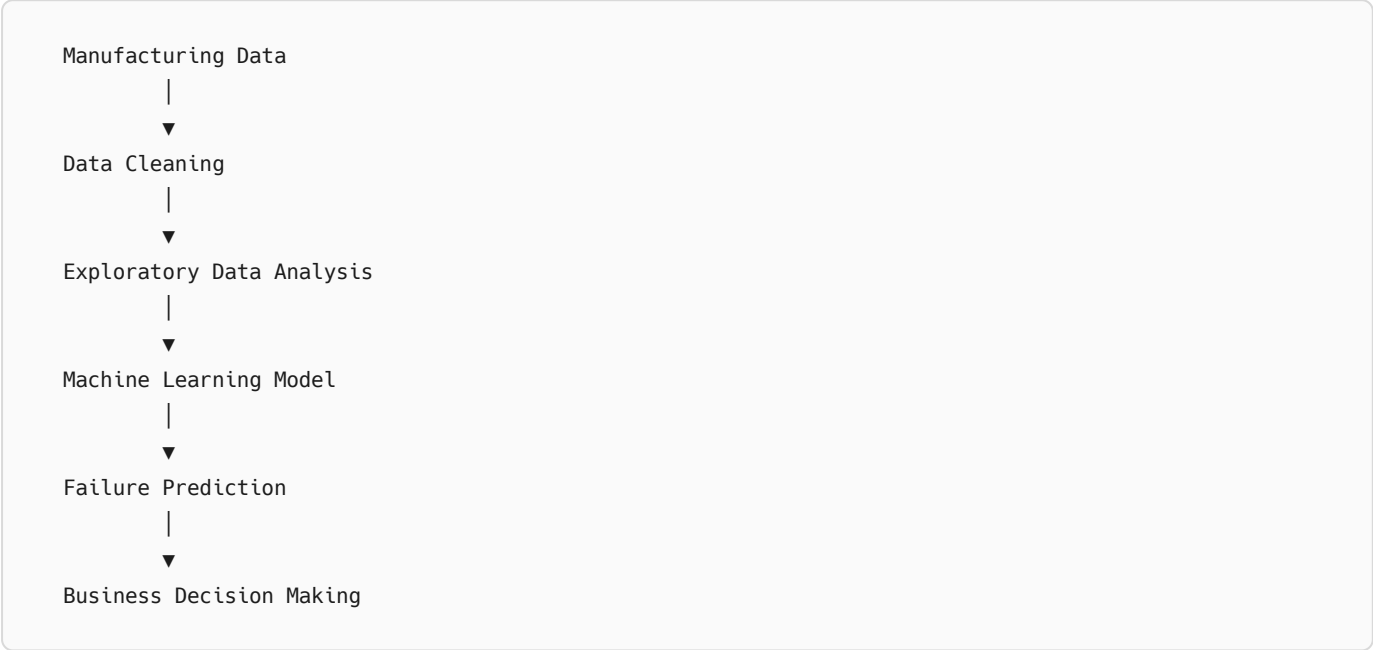
This project was developed to provide an end-to-end analytics solution capable of monitoring manufacturing operations, analyzing production quality, and predicting potential machine failures before they occur. By combining data analytics, exploratory data analysis (EDA), machine learning, and an interactive Streamlit dashboard, the solution supports proactive maintenance planning and data-driven decision-making.

Proposed Solution

The proposed solution follows a structured analytics workflow:

- 1. Collect and integrate manufacturing operational data.
 - 2. Perform data cleaning and preprocessing to ensure data quality.
 - 3. Conduct Exploratory Data Analysis (EDA) to identify operational trends and business insights.
 - 4. Develop a Random Forest Classifier to predict machine failures within the next seven days.
 - 5. Deploy the trained model through an interactive Streamlit application for real-time predictions and maintenance recommendations.
-

Figure : Project Workflow



Project Objectives

Project Overview

The primary objective of this project is to develop an end-to-end manufacturing analytics solution that enables organizations to improve production quality, reduce unexpected machine failures, and support proactive maintenance planning through data-driven decision-making. By integrating data analytics, exploratory data analysis (EDA), machine learning, and an interactive web application, the project transforms raw manufacturing data into meaningful business insights.

Business Objectives

The business objectives of this project are to:

- Improve manufacturing quality by identifying production trends and quality-related issues.
 - Reduce unexpected machine downtime through predictive maintenance.
 - Support preventive maintenance planning using machine learning predictions.
 - Enhance operational efficiency by monitoring machine performance.
 - Minimize maintenance costs associated with reactive maintenance.
 - Improve business decision-making through analytical dashboards and reports.
 - Increase equipment reliability and overall production productivity.
-

Analytical Objectives

The analytical objectives focus on extracting meaningful insights from manufacturing data.

- Perform comprehensive data cleaning and preprocessing.
 - Conduct Exploratory Data Analysis (EDA) to understand production and machine behavior.
 - Analyze sensor readings to identify abnormal operating conditions.
 - Study quality inspection data to evaluate manufacturing performance.
 - Explore maintenance records and failure history.
 - Identify relationships between operational variables and machine failures.
-

Machine Learning Objectives

The machine learning component aims to develop a reliable predictive maintenance solution.

- Build a classification model using the Random Forest algorithm.
 - Predict whether a machine is likely to fail within the next **7 days**.
 - Evaluate model performance using Accuracy, Precision, Recall, and F1-Score.
 - Generate failure probability scores for maintenance planning.
 - Support proactive maintenance recommendations based on prediction results.
-

Dashboard Objectives

The Streamlit application was developed to provide an interactive user experience.

The dashboard enables users to:

- Enter machine sensor readings.
- Predict machine failure instantly.

- View prediction confidence and failure probability.
 - Analyze machine input summary.
 - Receive maintenance recommendations.
 - Download prediction reports in CSV format.
-

Expected Business Outcomes

Successful implementation of this solution can help manufacturing organizations achieve:

- Reduced production downtime.
- Improved manufacturing quality.
- Better maintenance scheduling.
- Lower operational costs.
- Increased machine availability.
- Faster business decision-making.
- Higher productivity and operational efficiency.

Dataset Overview

Dataset Description

This project utilizes a comprehensive manufacturing dataset designed to simulate real-world industrial operations. The dataset combines information related to production, machine performance, sensor readings, quality inspections, maintenance activities, suppliers, downtime events, and machine failures. By integrating these business entities, the dataset provides a complete view of manufacturing operations and supports both analytical and predictive tasks.

The data was prepared to facilitate end-to-end data analysis, exploratory data analysis (EDA), and machine learning-based predictive maintenance. It reflects the operational environment of a modern manufacturing facility where multiple departments contribute to overall production efficiency and quality control.

Dataset Components

The project consists of multiple interconnected datasets, each representing a specific business function within the manufacturing process.

Machines	Machine information including type, model, installation date, and operating status.
Sensor Readings	Operational sensor measurements such as temperature, vibration, pressure, oil level, voltage, current, RPM, sound level, and machine health score.
Production	Daily production records, production line details, shift information, and output quantity.
Quality Inspections	Inspection results including defect counts, inspection status, and quality scores.
Maintenance Records	Scheduled and corrective maintenance history with maintenance costs and service details.
Failure Events	Historical machine failure records including failure type, severity, and failure date.

Downtime Events	Records of production interruptions caused by machine failures or maintenance activities.
Suppliers	Supplier information related to machine components and spare parts.

Key Features Used for Machine Learning

The predictive maintenance model uses the following operational features:

Temperature (°C)	Machine operating temperature
Vibration (mm/s)	Vibration level of the machine
Pressure (bar)	Operating pressure
Oil Level (%)	Lubrication level
Voltage (V)	Electrical voltage supplied
Current (A)	Current consumption
RPM	Rotational speed
Sound Level (dB)	Machine noise level
Machine Health Score	Overall health indicator

Target Variable

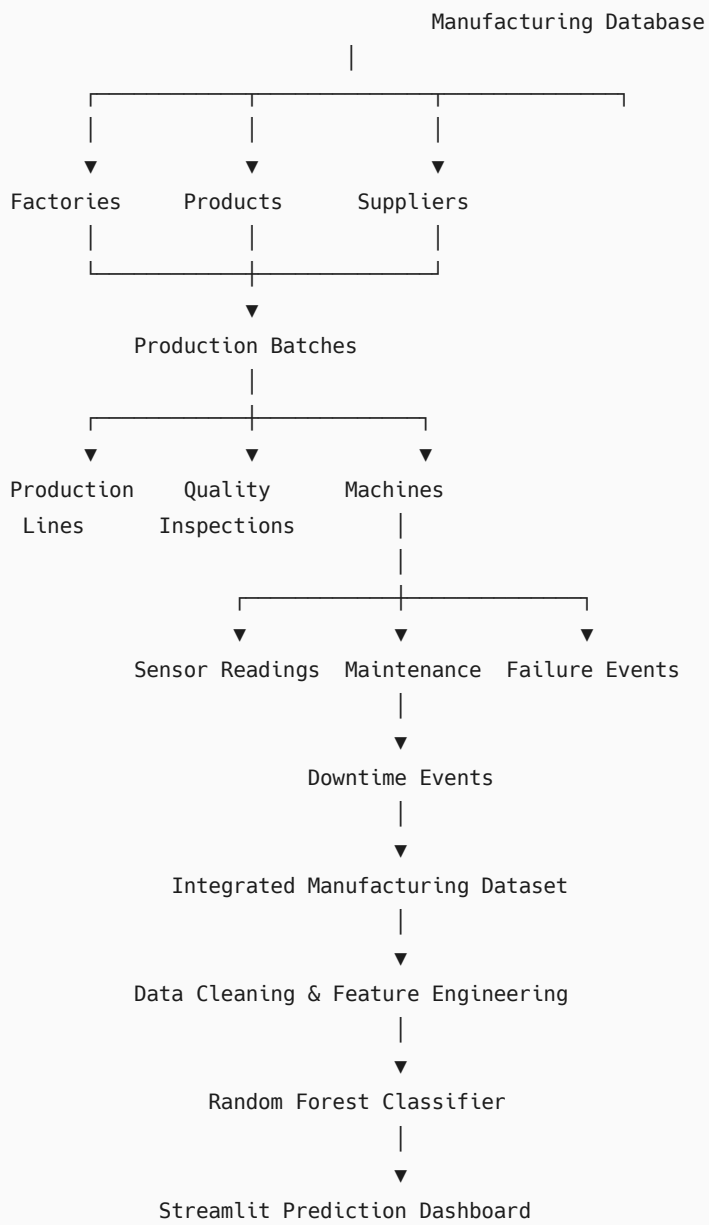
- **Machine Failure (Yes / No)** — Indicates whether the machine is predicted to fail within the next **7 days**.

Business Domains Covered

This dataset supports analysis across multiple manufacturing domains:

- Manufacturing Operations
- Quality Control
- Predictive Maintenance
- Machine Performance Analysis
- Production Monitoring
- Sensor Analytics
- Downtime Analysis
- Operational Efficiency
- Business Intelligence

 **Figure: Dataset Architecture**



Data Cleaning & Preprocessing

Overview

Raw manufacturing data collected from operational systems often contains inconsistencies, missing information, duplicate records, incorrect data types, and invalid values. Before performing business analysis or developing machine learning models, it is essential to ensure that the dataset is accurate, complete, and reliable.

A structured data cleaning and preprocessing workflow was implemented to improve data quality and prepare the manufacturing database for exploratory data analysis (EDA) and predictive maintenance modeling.

	Dataset	Rows	Columns	Missing Values	Duplicate Rows	Memory usage (MB)
0	Factories	10	9	0	0	0.00
1	Production Lines	30	7	0	0	0.01
2	Machines	600	17	0	0	0.33
3	Suppliers	210	8	0	0	0.07
4	Products	150	10	0	0	0.06
5	Operators	1000	9	0	0	0.35
6	Production Batches	60000	24	450	0	41.54
7	Sensor Readings	285000	17	278485	0	125.22
8	Maintenance Work Orders	12000	16	9000	0	8.34
9	Quality Inspections	30000	18	192	0	22.11
10	Downtime Events	8000	13	3704	0	4.54
11	Failure Events	3000	11	0	0	1.40

Figure1 : Overview of Dataset before Data Cleaning .

Data Cleaning Activities

Missing Value Analysis	Identify and handle incomplete records.
Duplicate Detection	Remove duplicate records to prevent biased analysis.
Data Type Validation	Ensure numerical, categorical, and date fields use appropriate data types.
Outlier Detection	Identify abnormal observations using statistical techniques.
Consistency Checks	Validate relationships between operational variables.
Data Standardization	Standardize formats for reliable analysis.

Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the manufacturing dataset. The objective was to uncover operational trends, identify performance patterns, detect anomalies, and generate actionable business insights before building the predictive maintenance model.

The analysis focused on multiple aspects of manufacturing operations, including production performance, machine health, quality inspections, sensor behavior, maintenance activities, downtime events, and failure patterns. These insights provided a strong analytical foundation for both business decision-making and machine learning.

TOTAL PRODUCTION BY FACTORIES

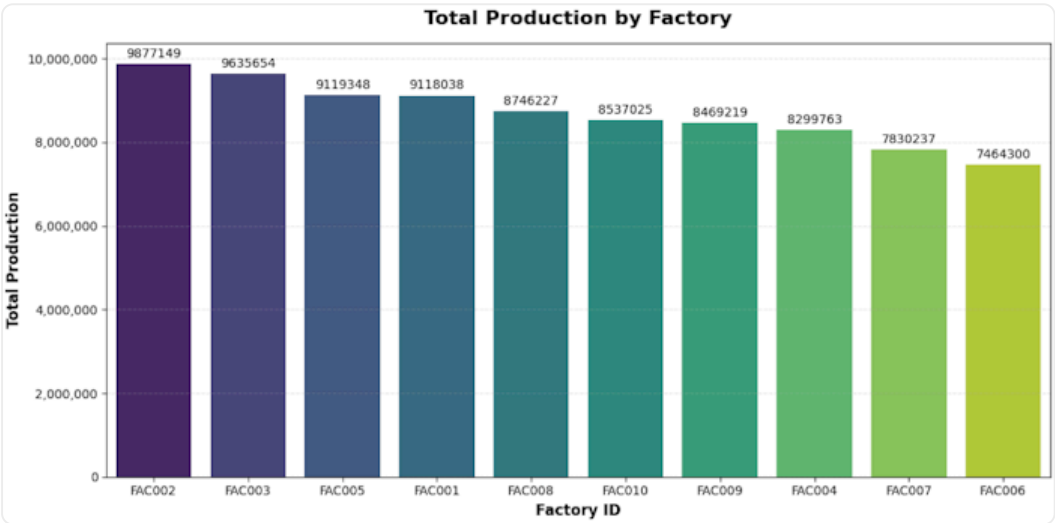


FIGURE 2 : PRODUCTION BY FACTORIES

TOTAL MACHINES BY FACTORY

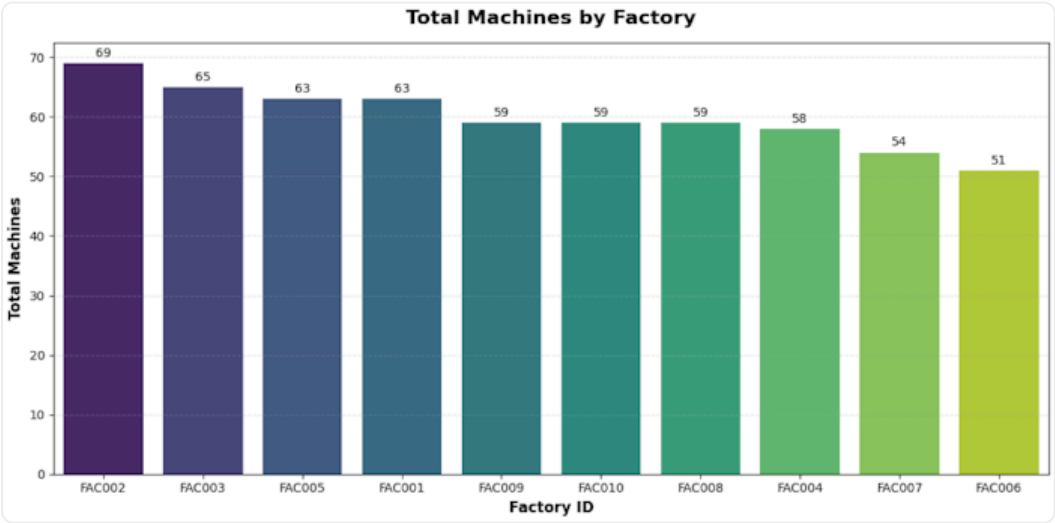


FIGURE 3 : TOTAL MACHINES BY FACTORY

Observation

The analysis indicates a positive relationship between the number of machines and total production across factories. Factories with a larger machine base generally achieved higher production output. For example, **FAC002**, which has the

highest number of machines (**69**), also recorded the highest production of approximately **9.88 million units**. Similarly, **FAC003** and **FAC005**, with **65** and **63** machines respectively, ranked among the top producers. In contrast, **FAC006**, which operates with the fewest machines (**51**), reported the lowest production of approximately **7.46 million units**. This trend suggests that machine availability plays an important role in determining production capacity, although operational efficiency, maintenance practices, and machine utilization also influence overall output.

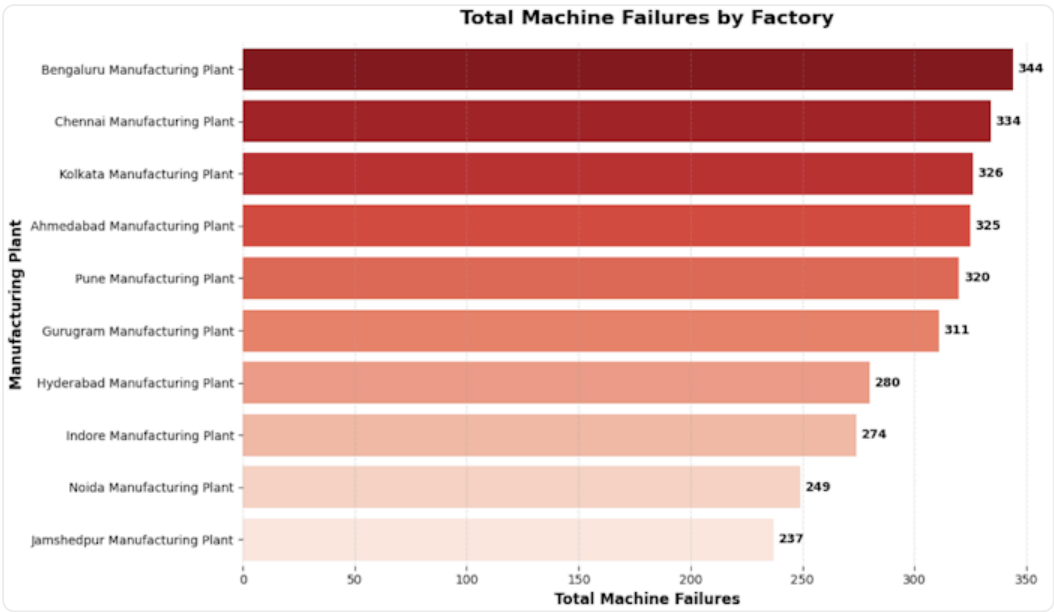


FIGURE 4: Total Machine Failure by Factory

DOWNTIME LOSS BY FACTORY

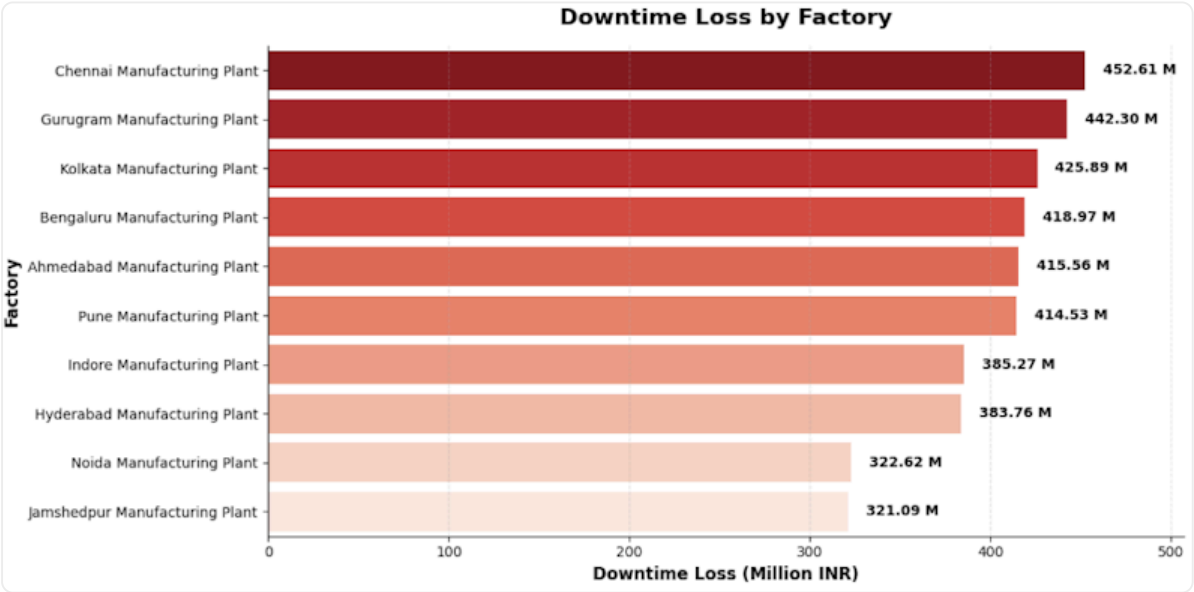


FIGURE 7: Downtime Loss By factory

Observation

The analysis indicates a strong relationship between **machine failures** and **downtime losses** across manufacturing plants. **Chennai Manufacturing Plant** recorded the highest downtime loss of approximately **₹452.61 million**, followed by **Gurugram (₹442.30 million)** and **Kolkata (₹425.89 million)**. Notably, plants such as **Kolkata** and **Bengaluru** also ranked among those with the highest machine failure counts, suggesting that frequent equipment failures contribute significantly to production downtime and financial losses. In contrast, **Noida** and **Jamshedpur** reported the lowest downtime losses, indicating comparatively better operational stability and equipment reliability.

Business Insight

The findings highlight that reducing machine failures can have a direct impact on minimizing downtime-related financial losses. Manufacturing plants with high downtime costs should be prioritized for **predictive maintenance**, **root cause analysis**, and **continuous monitoring of critical sensor parameters**. By identifying high-risk machines before failure, organizations can reduce production interruptions, lower maintenance expenses, and improve overall operational efficiency and profitability.

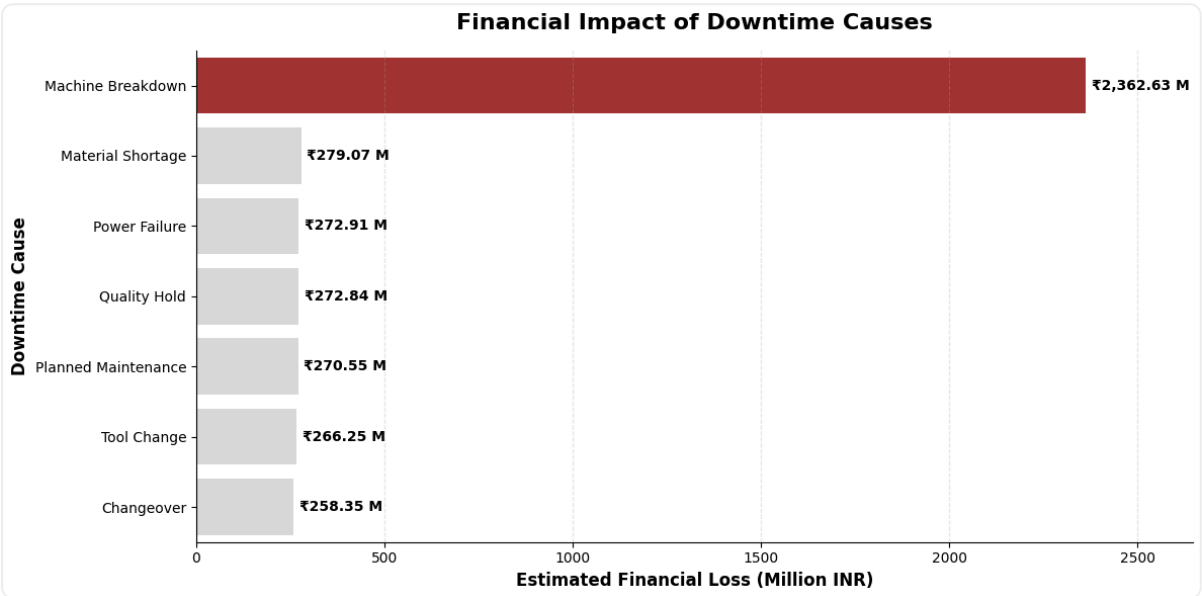


FIGURE 5: Financial Impact of Downtime Causes

Observation

The analysis shows that **Machine Breakdown** is the most significant contributor to financial losses, resulting in an estimated downtime cost of approximately **₹2,362.63 million**, which is substantially higher than all other downtime causes. In comparison, **Material Shortage (₹279.07 million)**, **Power Failure (₹272.91 million)**, **Quality Hold (₹272.84 million)**, **Planned Maintenance (₹270.55 million)**, **Tool Change (₹266.25 million)**, and **Changeover (₹258.35 million)** contribute relatively similar and significantly lower financial losses. This indicates that unexpected machine breakdowns have a far greater economic impact than routine operational interruptions.

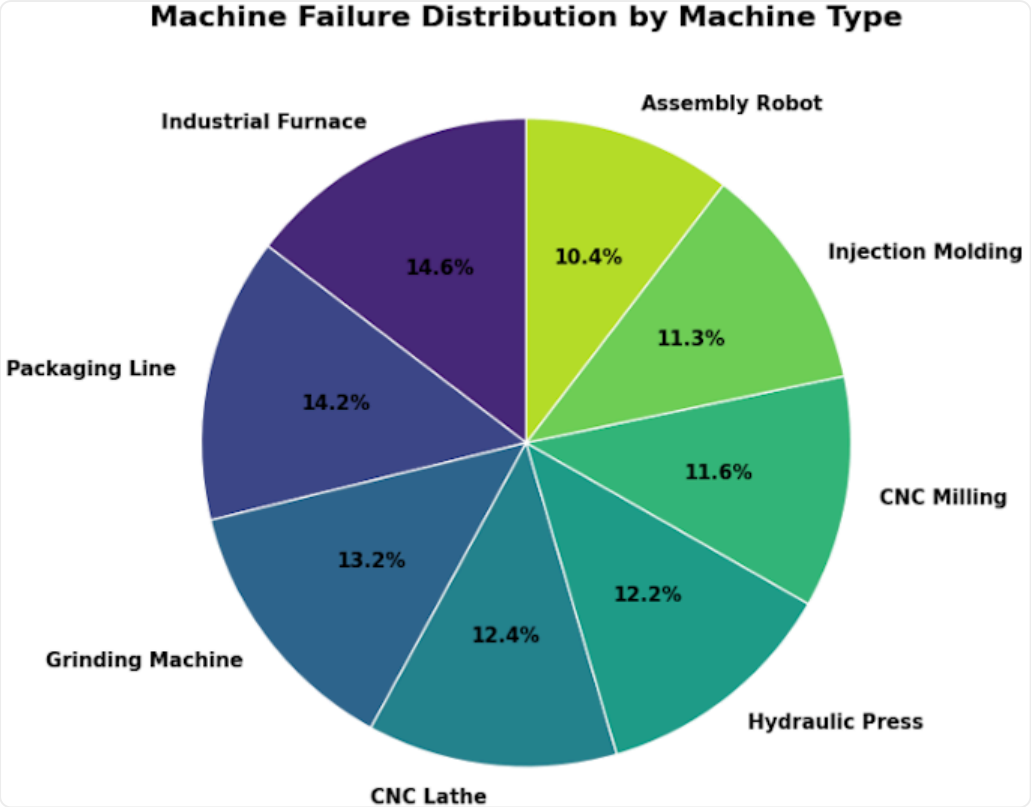


FIGURE 5: Machine Type Failure

Observation

The machine failure distribution indicates that **Industrial Furnaces** account for the highest proportion of machine failures (**14.6%**), followed closely by **Packaging Lines** (**14.2%**) and **Grinding Machines** (**13.2%**). On the other hand, **Assembly Robots** (**10.4%**), **Injection Molding Machines** (**11.3%**), and **CNC Milling Machines** (**11.6%**) recorded comparatively lower failure rates. The relatively balanced distribution across machine types suggests that failures are not concentrated in a single category; however, Industrial Furnaces and Packaging Lines exhibit a higher failure frequency and therefore represent the most critical equipment from a maintenance perspective.

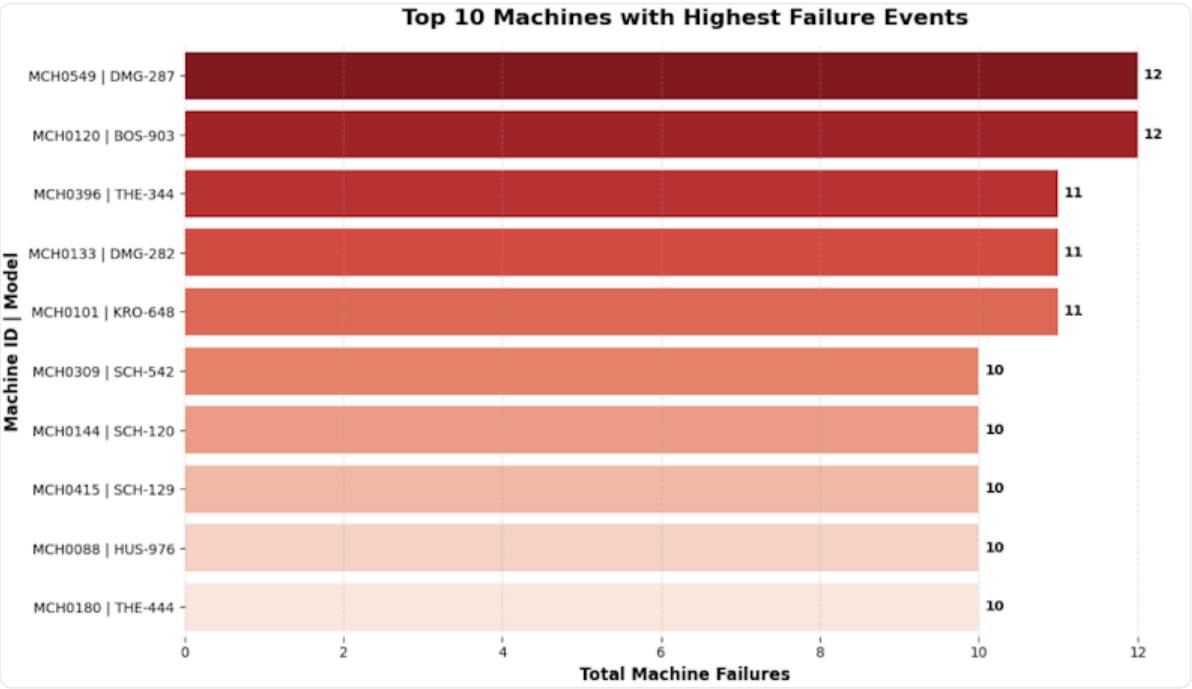


FIGURE 6: TOP 10 Machines With Highest Failure

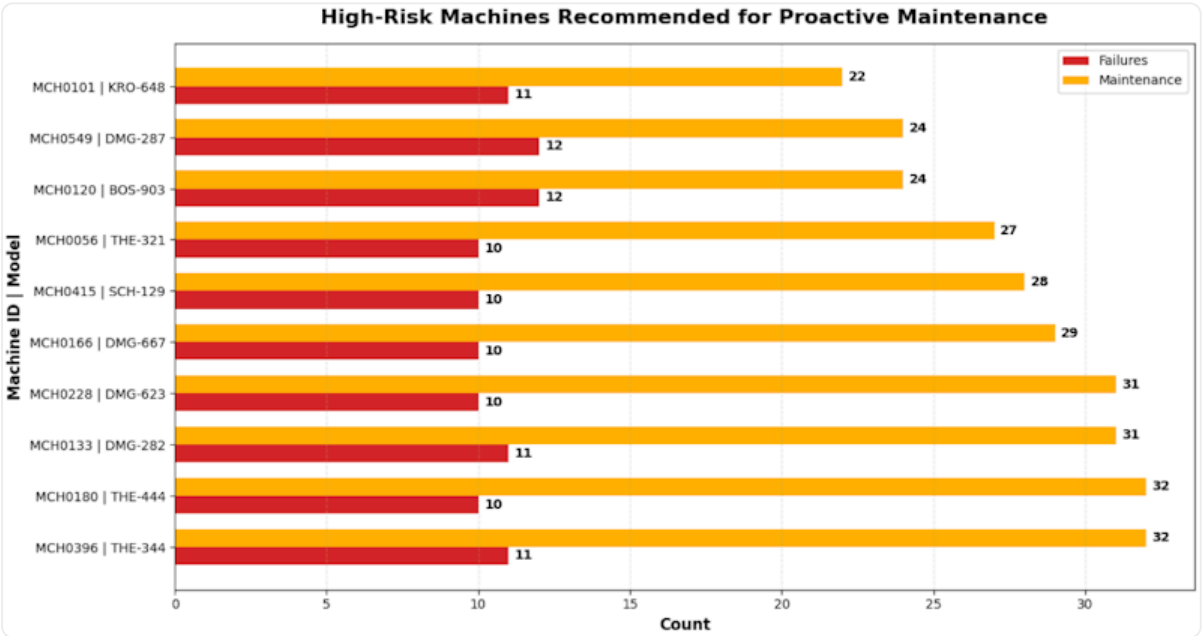
Observation

The analysis identifies the **top 10 machines with the highest number of failure events**, highlighting equipment that requires immediate maintenance attention. **Machine MCH0549 (DMG-287)** and **MCH0120 (BOS-903)** recorded the highest number of failures, with **12 failure events** each. They were followed by **MCH0396 (THE-344)**, **MCH0133 (DMG-282)**, and **MCH0101 (KRO-648)**, each experiencing **11 failure events**. The remaining machines reported **10 failure events**, indicating that these assets consistently experience reliability issues. The concentration of repeated failures in these machines suggests recurring mechanical, operational, or maintenance-related problems rather than isolated incidents.

Business Insight

Machines with the highest failure frequency should be classified as **high-priority maintenance assets**. Performing root cause analysis, reviewing historical maintenance records, and continuously monitoring critical sensor parameters such as **temperature, vibration, pressure, and machine health score** can help reduce recurring failures. Prioritizing these machines for preventive and predictive maintenance will improve equipment reliability, reduce unplanned downtime, and enhance overall production efficiency.

Figure: High-Risk Machines Recommended for Proactive Maintenance



Observation

The analysis identifies machines that exhibit both **high failure frequency** and **high maintenance activity**, making them the most critical assets for proactive maintenance. **MCH0396 (THE-344)** and **MCH0180 (THE-444)** recorded the highest maintenance counts (**32 work orders**) while also experiencing **11** and **10 failure events**, respectively. Similarly, **MCH0133 (DMG-282)** and **MCH0228 (DMG-623)** required **31 maintenance work orders**, indicating recurring maintenance demand alongside repeated failures. These patterns suggest that these machines are operating under higher operational stress or require further investigation to address recurring reliability issues.

Business Insight

These machines should be classified as **high-priority assets** within the predictive maintenance program. Continuous monitoring of critical sensor parameters, preventive servicing, and detailed root cause analysis can help reduce recurring failures and maintenance interventions. Prioritizing these machines for predictive maintenance will improve equipment reliability, reduce unexpected downtime, optimize maintenance resources, and enhance overall production efficiency.

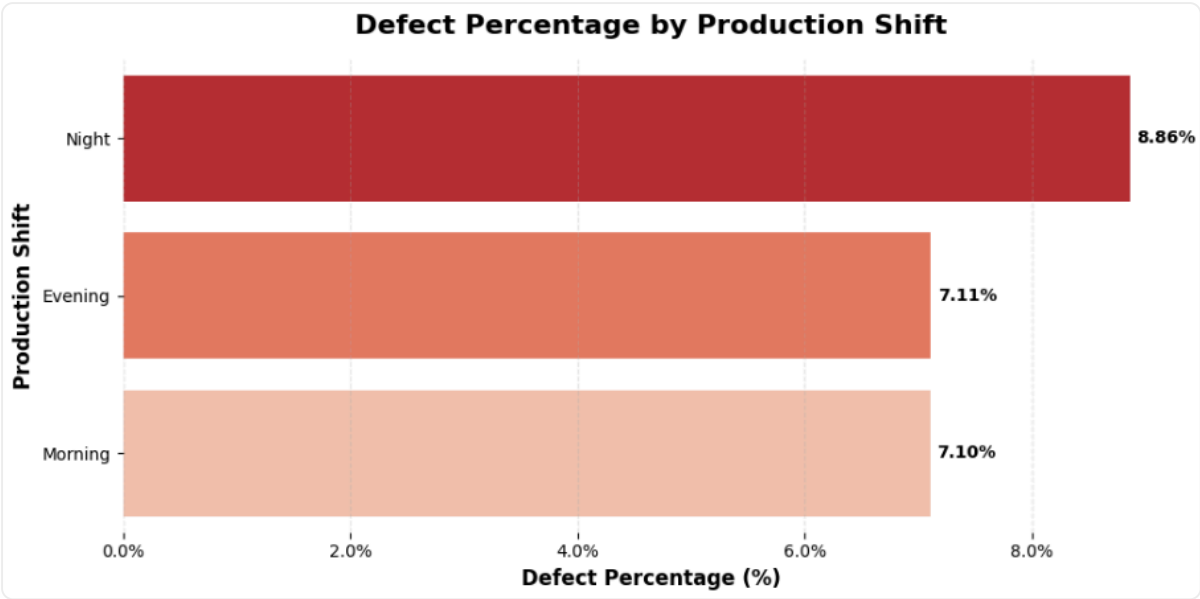


FIGURE 8: Defection by Production Shift

Observation

The analysis indicates that the **Night Shift** recorded the highest defect percentage at **8.86%**, significantly higher than both the **Evening Shift (7.11%)** and **Morning Shift (7.10%)**. The Morning and Evening shifts show nearly identical defect rates, while the Night Shift demonstrates a noticeable increase in product defects. This trend suggests that production quality varies across shifts and that operational conditions during the Night Shift may be contributing to a higher occurrence of defective products.

Business Insight

The consistently higher defect rate during the **Night Shift** indicates the need for targeted quality improvement initiatives. Manufacturing management should investigate factors such as operator fatigue, reduced supervision, machine maintenance schedules, and production workload during night operations. Implementing additional quality inspections, operator training, and real-time monitoring during the Night Shift can help reduce defect rates, improve product quality, and minimize production losses.

Predictive Maintenance Dashboard: Average Sensor Readings by Failure Status

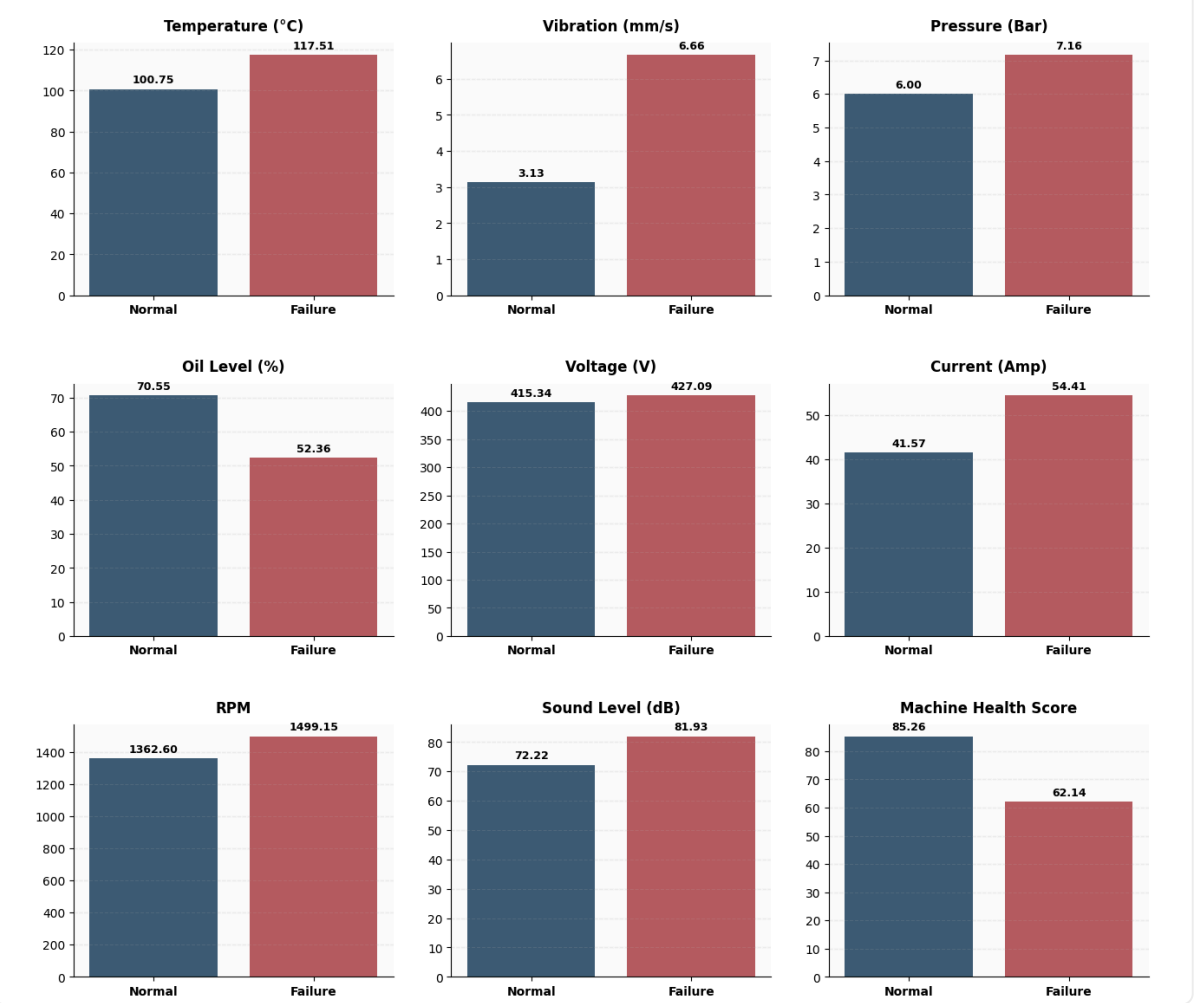


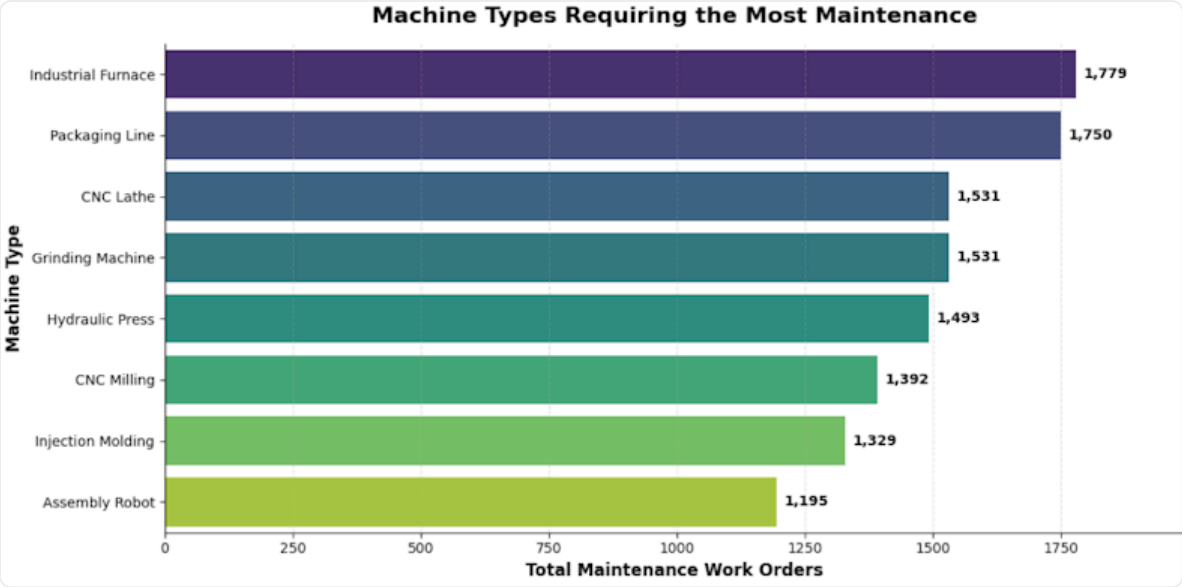
FIGURE 9 : Sensor Status by Reading

Observation

The comparative analysis of sensor readings reveals clear differences between **normal** and **failure** operating conditions. Machines that experienced failures showed significantly higher **temperature (117.51°C)**, **vibration (6.66 mm/s)**, **pressure (7.16 bar)**, **voltage (427.09 V)**, **current (54.41 A)**, **RPM (1499.15)**, and **sound level (81.93 dB)** compared to machines operating under normal conditions. In contrast, **oil level** decreased substantially from **70.55%** to **52.36%**, while the **machine health score** dropped sharply from **85.26** to **62.14** during failure conditions. These consistent changes indicate that deteriorating sensor values are strong indicators of machine failure.

Business Insight

The results demonstrate that sensor data can provide early warning signs of equipment deterioration before a machine fails. Continuous monitoring of critical parameters such as **temperature, vibration, pressure, oil level, and machine health score** enables maintenance teams to identify abnormal operating conditions and schedule preventive maintenance proactively. Implementing threshold-based alerts for these sensors can reduce unexpected breakdowns, minimize downtime, improve equipment reliability, and lower maintenance costs.

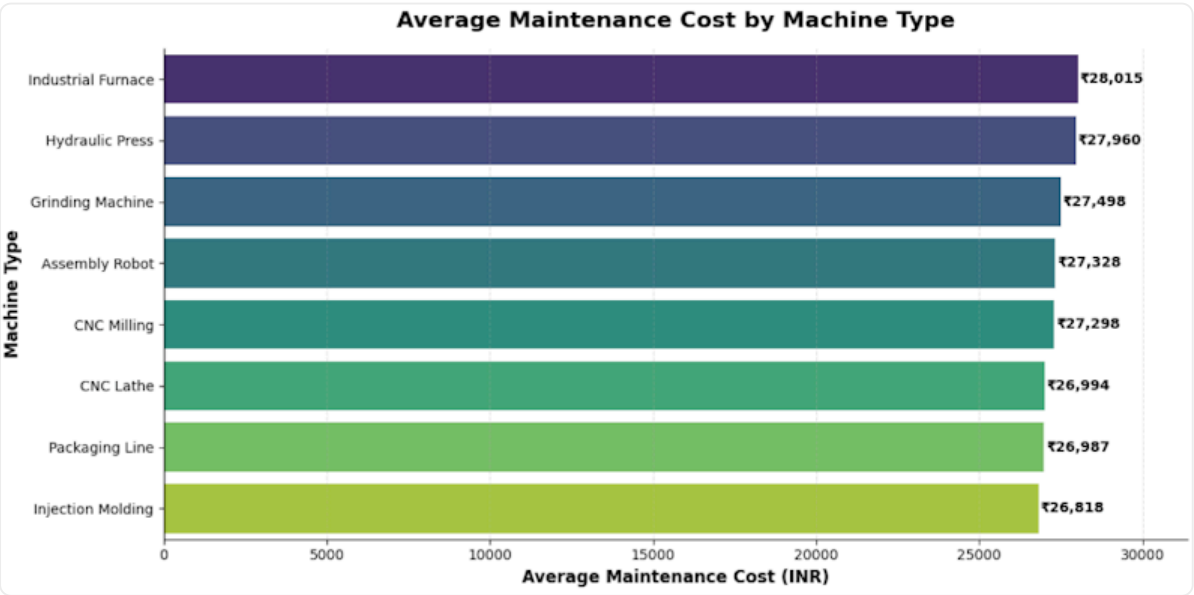


Observation

The analysis reveals that **Industrial Furnaces** require the highest number of maintenance work orders (**1,779**), followed by **Packaging Lines (1,750)** and **CNC Lathes (1,531)**. These machine types were also identified among those with the highest machine failure rates in the previous analysis, indicating a strong relationship between frequent equipment failures and increased maintenance requirements. In contrast, **Assembly Robots** required the fewest maintenance work orders (**1,195**), suggesting comparatively better operational reliability and lower maintenance demand.

Business Insight

The findings indicate that **Industrial Furnaces** and **Packaging Lines** should be prioritized for preventive and predictive maintenance programs, as they exhibit both higher failure frequencies and greater maintenance requirements. Regular monitoring of critical sensor parameters, timely servicing, and root cause analysis can reduce recurring maintenance activities, improve equipment reliability, and minimize production downtime. Optimizing maintenance schedules for these high-risk machine types can significantly improve operational efficiency while reducing maintenance costs.

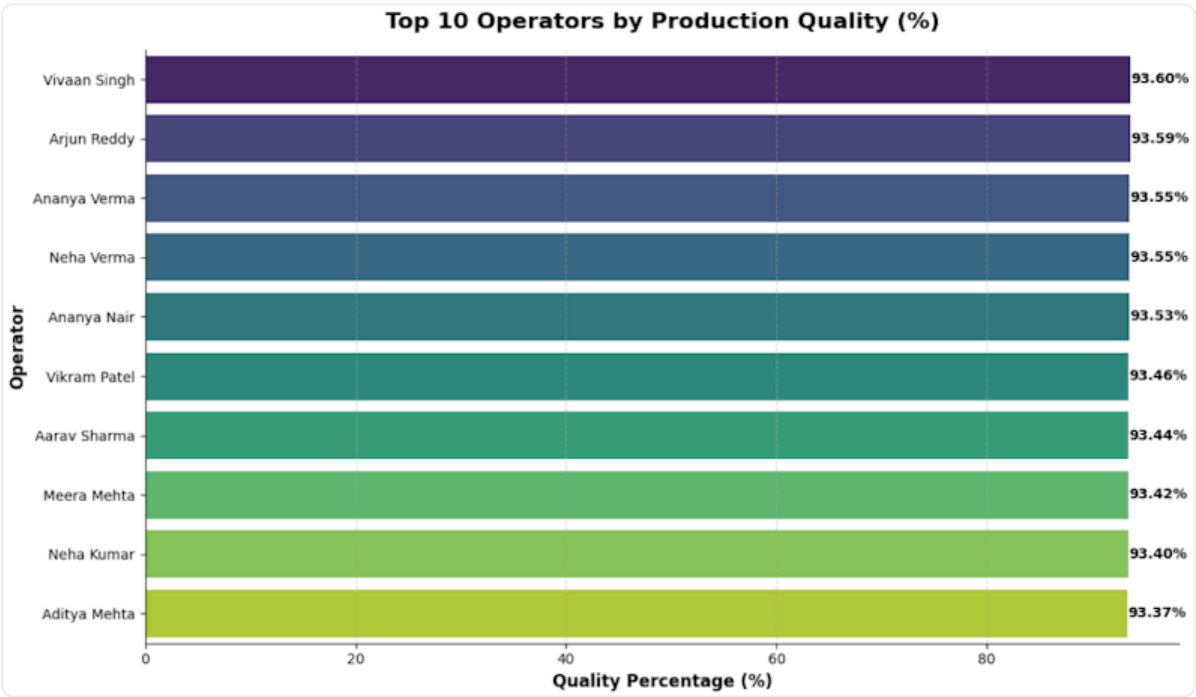


Observation

The analysis shows that **Industrial Furnaces** incur the highest average maintenance cost at approximately **₹28,015** per maintenance activity, followed closely by **Hydraulic Presses (₹27,960)** and **Grinding Machines (₹27,498)**. Although the difference in average maintenance cost across machine types is relatively small, **Industrial Furnaces** stand out because they also recorded the highest number of maintenance work orders and one of the highest machine failure rates in the previous analyses. This indicates that Industrial Furnaces not only require frequent maintenance but also involve higher maintenance expenditure, making them one of the most resource-intensive assets in the manufacturing environment.

Business Insight

The combination of **high maintenance frequency** and **high average maintenance cost** makes **Industrial Furnaces** a critical area for optimization. Manufacturing organizations should prioritize predictive maintenance, continuous sensor monitoring, and root cause analysis for these machines to reduce recurring maintenance activities and maintenance expenses. Optimizing maintenance schedules for high-cost equipment can improve machine reliability, reduce operational costs, and increase overall manufacturing efficiency.



Observation

The analysis highlights the operators who consistently achieved the highest production quality. **Vivaan Singh** recorded the highest production quality at **93.60%**, closely followed by **Arjun Reddy (93.59%)**, **Ananya Verma (93.55%)**, and **Neha Verma (93.55%)**. The quality percentages among the top operators are consistently above **93%**, indicating strong adherence to manufacturing standards and operational best practices.

Business Insight

High-performing operators can serve as benchmarks for production excellence. Their working practices, adherence to standard operating procedures (SOPs), and production techniques should be analyzed and incorporated into operator training programs. Sharing best practices across shifts can improve overall production quality, reduce defect rates, and enhance manufacturing consistency.

Machine Learning Implementation

Introduction

To support predictive maintenance, a machine learning model was developed to predict whether a machine is likely to fail within the next **seven days** based on operational sensor readings. The objective was to identify potential equipment failures before they occur, enabling maintenance teams to take preventive actions and reduce unplanned downtime.

A **Random Forest Classifier** was selected for this project due to its ability to handle complex relationships between sensor variables, robustness against overfitting, and strong performance on classification tasks involving manufacturing data.

Problem Statement

Unexpected machine failures can interrupt production, increase maintenance costs, and reduce operational efficiency. Traditional reactive maintenance approaches address failures only after they occur, resulting in higher downtime and financial losses.

This project addresses this challenge by developing a predictive model that analyzes machine sensor data to estimate the likelihood of machine failure within the next seven days, enabling proactive maintenance planning.

Machine Learning Objective

The primary objectives of the machine learning model are:

- Predict machine failures before they occur.
- Identify high-risk machines using sensor data.
- Support predictive maintenance planning.
- Reduce unexpected production downtime.
- Improve machine reliability and operational efficiency.
- Enable data-driven maintenance decisions.

Target Variable

The target variable used for model training is:

failure_within_7_days	Indicates whether a machine is expected to fail within the next seven days.
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Type: **Binary Classification**

- 0 → No Failure
- 1 → Failure

9.5 Input Features

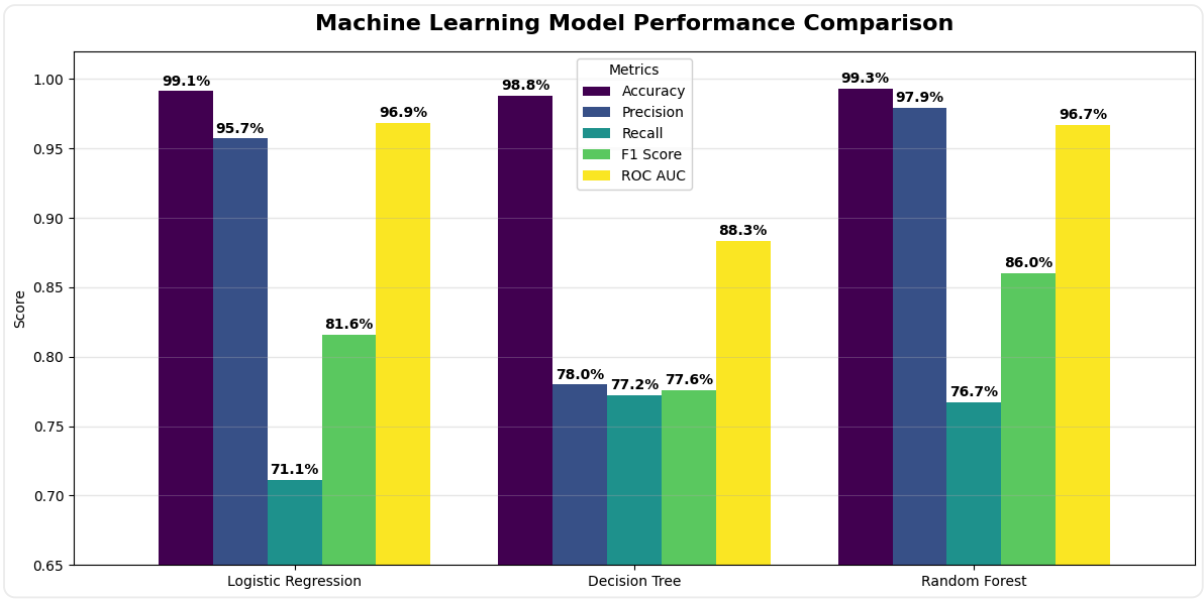
The model uses the following sensor measurements as input features:

Temperature (°C)	Machine operating temperature
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Vibration (mm/s)	Machine vibration level
Pressure (Bar)	Operating pressure
Oil Level (%)	Lubrication level
Voltage (V)	Supply voltage
Current (Amp)	Current consumption
RPM	Machine rotational speed
Sound Level (dB)	Noise level
Machine Health Score	Overall machine condition

Machine Learning Implementation

Machine Learning Model Performance Comparison



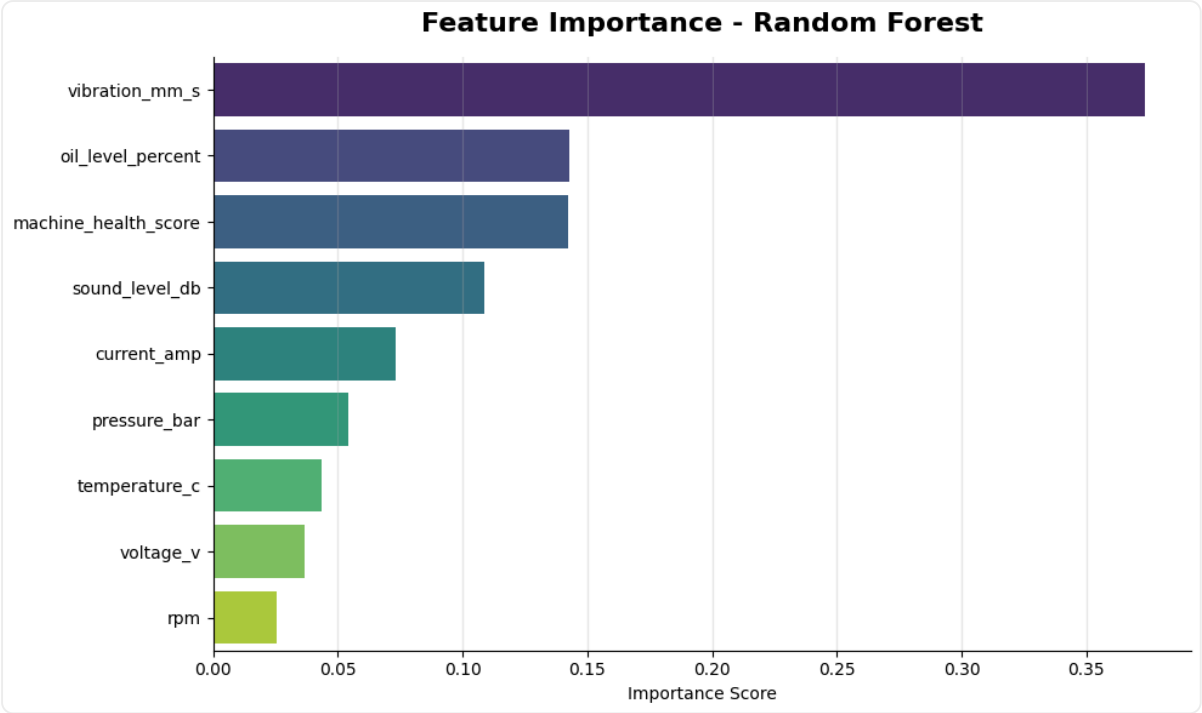
Observation

Three classification models—**Logistic Regression**, **Decision Tree**, and **Random Forest**—were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Among them, the **Random Forest Classifier** achieved the highest overall performance with **99.3% Accuracy**, **97.9% Precision**, **76.7% Recall**, **86.0% F1-Score**, and **96.7% ROC-AUC**. These results indicate that Random Forest provides the most reliable balance between predictive accuracy and generalization for this manufacturing dataset.

Business Insight

The Random Forest model was selected for deployment because it consistently outperformed the other models. Its high predictive performance enables the early identification of high-risk machines, helping maintenance teams reduce unexpected failures, minimize downtime, and improve operational efficiency.

Figure 16: Feature Importance – Random Forest



Observation

The feature importance analysis shows that **Vibration (mm/s)** is the most influential predictor of machine failure, followed by **Oil Level (%)** and **Machine Health Score**. Features such as **Sound Level**, **Current**, and **Pressure** also contribute significantly, while **RPM**, **Voltage**, and **Temperature** have comparatively lower importance in the model.

Business Insight

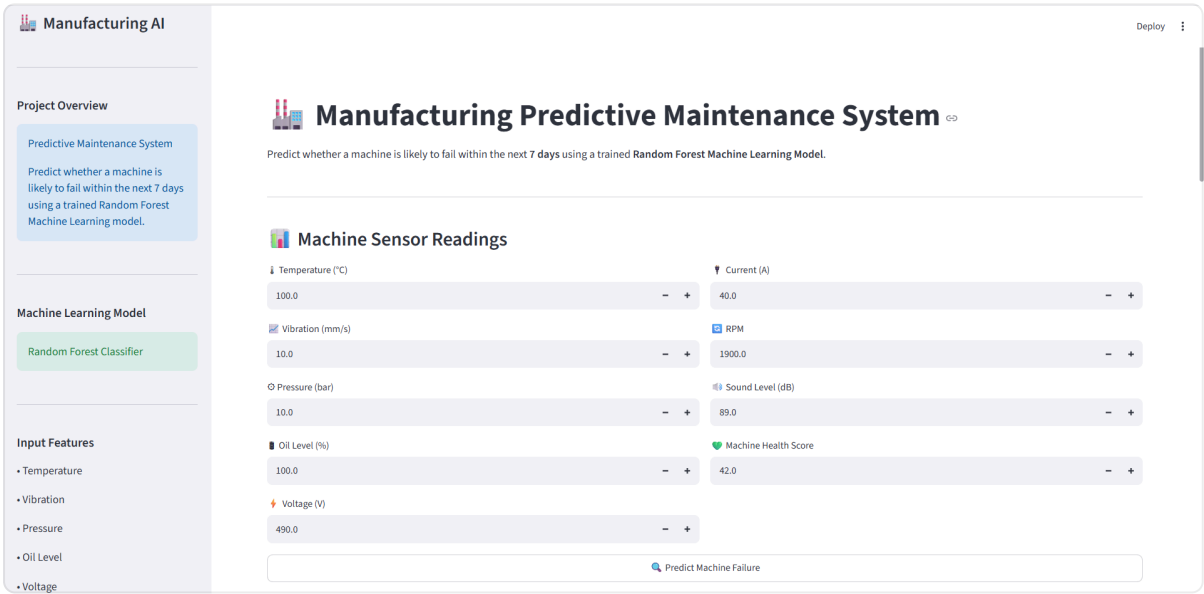
The results suggest that maintenance teams should prioritize monitoring **vibration**, **lubrication (oil level)**, and **overall machine health** to detect potential failures at an early stage. Focusing on these critical indicators can improve predictive maintenance planning, reduce unexpected breakdowns, and optimize maintenance resources.

Streamlit Dashboard

Dashboard Overview

The trained **Random Forest Classifier** was deployed as an interactive **Streamlit web application** to provide real-time machine failure predictions. The dashboard enables users to enter machine sensor readings, predict the likelihood of machine failure within the next seven days, and receive maintenance recommendations through a simple and user-friendly interface.

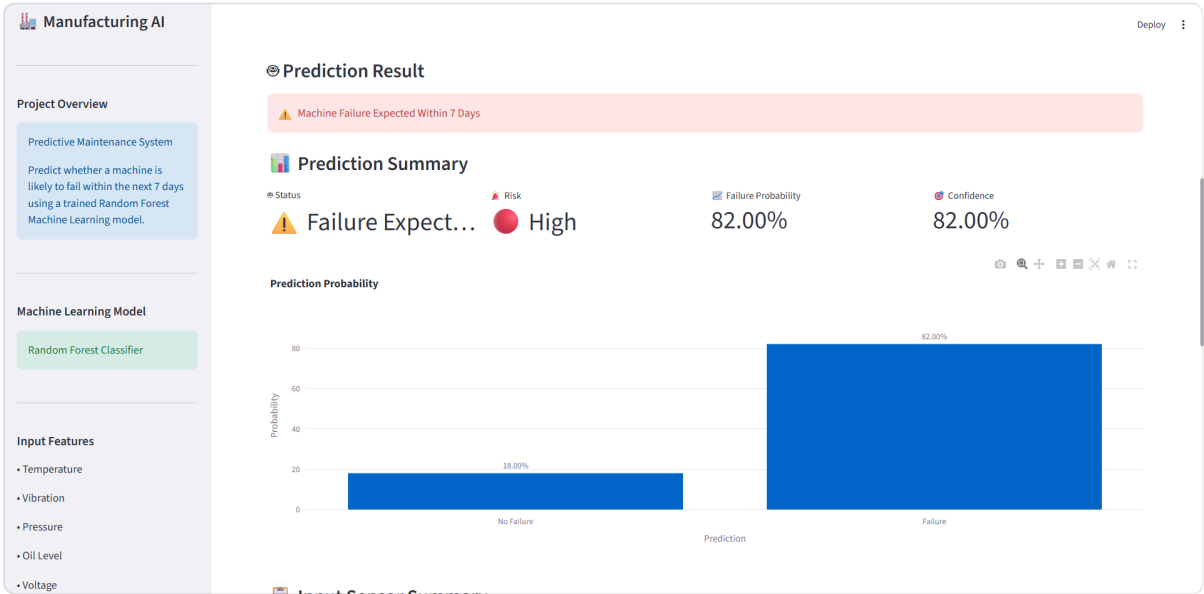
Figure 17: Dashboard Home Page



Description

The home page provides an intuitive interface where users can enter machine sensor values such as temperature, vibration, pressure, oil level, voltage, current, RPM, sound level, and machine health score. Once the values are entered, the model generates a machine failure prediction.

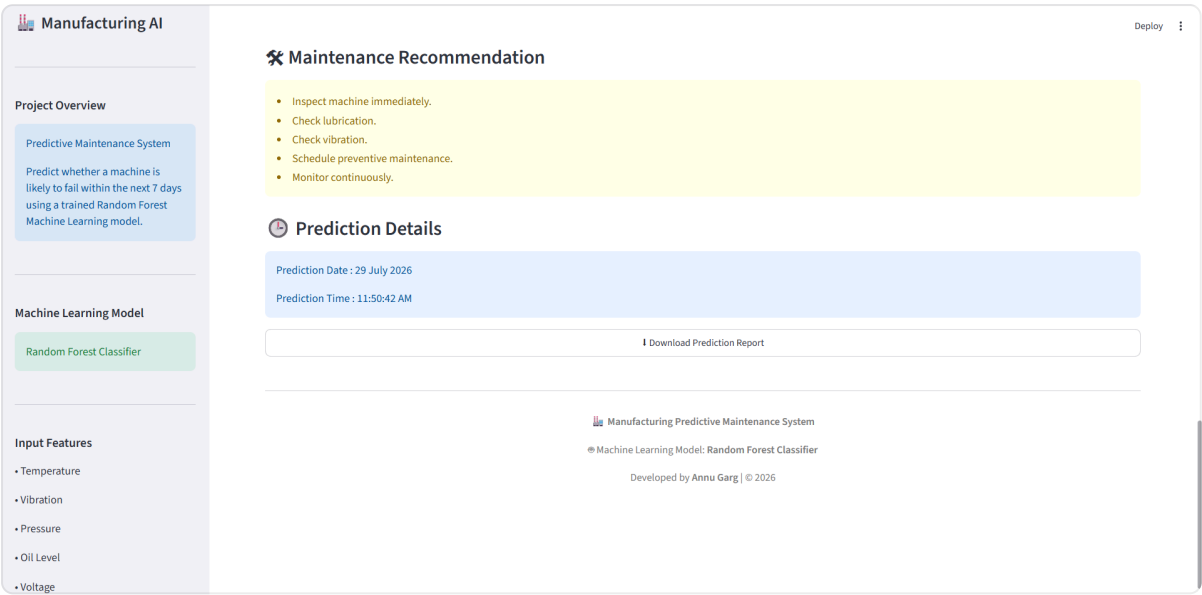
Figure 18: Prediction Result



Description

The dashboard displays whether the machine is at **High Risk** or **Low Risk** of failure along with the predicted failure probability, enabling maintenance teams to quickly identify machines that require immediate attention.

Figure 19: Maintenance Recommendation



Description

Based on the prediction result, the application provides maintenance recommendations to help engineers plan preventive maintenance and reduce the likelihood of unexpected machine breakdowns.

Users can download the prediction results as a CSV or PDF report for documentation, reporting, and future analysis.

Business Value

The Streamlit dashboard transforms the trained machine learning model into a practical decision-support tool. By combining real-time predictions with maintenance recommendations and downloadable reports, the application helps organizations reduce unplanned downtime, improve maintenance planning, and support data-driven operational decisions.

Business Recommendations

Based on the findings from Exploratory Data Analysis (EDA) and the Machine Learning model, the following recommendations are proposed to improve manufacturing efficiency and reduce machine failures:

- Implement predictive maintenance for machines identified as high risk by the Random Forest model.
- Continuously monitor critical sensor parameters, especially **vibration**, **oil level**, and **machine health score**, as they have the greatest impact on machine failure prediction.
- Prioritize maintenance for machine types with higher failure frequencies, such as **Industrial Furnaces** and **Packaging Lines**.
- Strengthen quality control during the **Night Shift**, where defect rates were observed to be higher.
- Reduce downtime caused by machine breakdowns through preventive maintenance and regular equipment inspections.
- Use the Streamlit dashboard as a decision-support tool for real-time monitoring and maintenance planning.

Conclusion

This project successfully developed an end-to-end **Manufacturing Quality Control and Predictive Maintenance** solution by integrating data cleaning, exploratory data analysis, machine learning, and an interactive Streamlit application. The analysis identified key operational patterns, major causes of downtime, high-risk machines, and the sensor variables most strongly associated with machine failures.

The **Random Forest Classifier** demonstrated strong predictive performance and was successfully deployed through a Streamlit dashboard, enabling real-time failure prediction and maintenance recommendations. Overall, the project shows how data-driven decision-making can improve equipment reliability, reduce unplanned downtime, and enhance operational efficiency.

References

The following tools, libraries, and technologies were used during the development of this project:

Programming & Analysis

- Python 3.x
- Pandas
- NumPy
- Matplotlib
- Plotly
- Scikit-learn
- Joblib

Database

- MySQL

Machine Learning

- Random Forest Classifier
- Logistic Regression
- Decision Tree

Deployment

- Streamlit

Development Environment

- Jupyter Notebook
- Visual Studio Code

Thank You

| Thank You for Reviewing This Project

This project demonstrates an end-to-end manufacturing analytics solution that integrates **Data Cleaning, SQL, Exploratory Data Analysis (EDA), Machine Learning, and Streamlit Deployment** to support predictive maintenance and improve manufacturing decision-making.

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